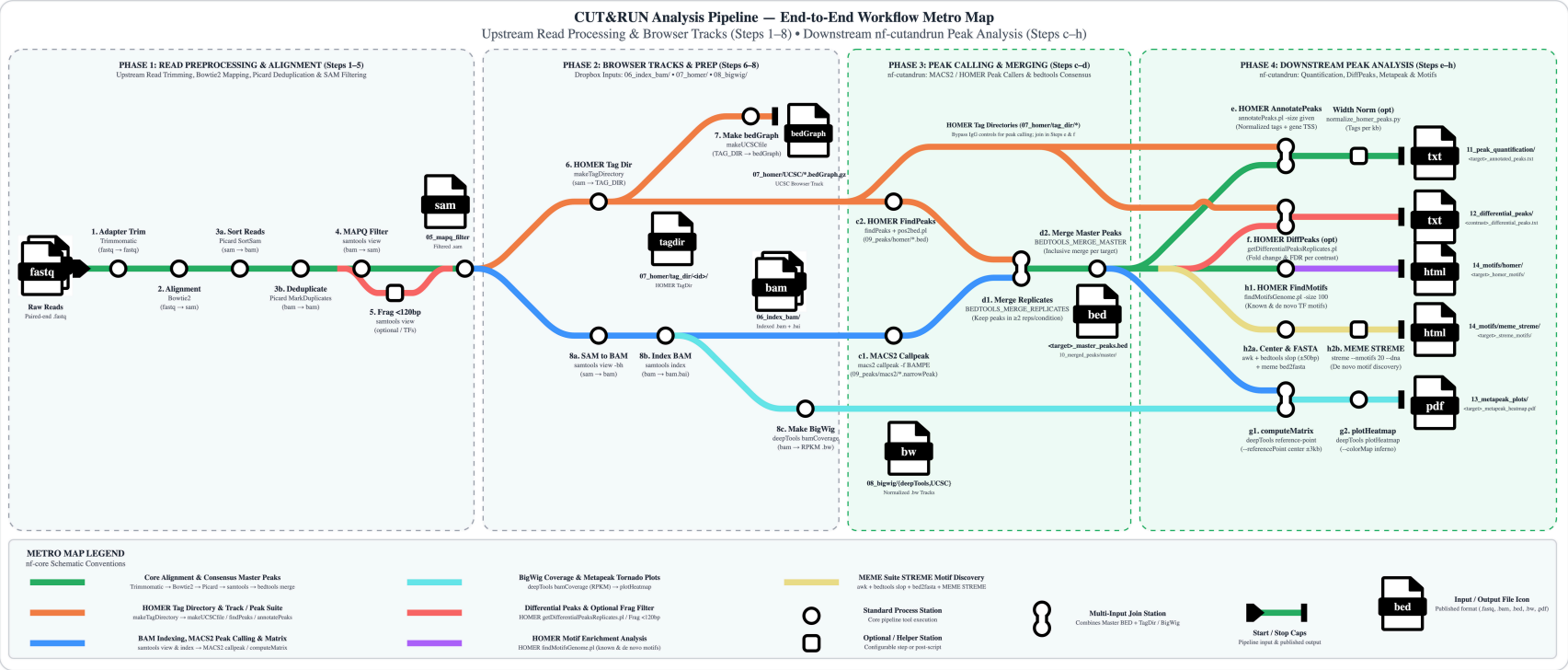
1. CUT&RUN Analysis Pipeline — End-to-End Metro Map

High-Level Architecture: From Raw Paired-End FASTQs to Downstream Peak & Motif Discovery



Phases 1–2: Upstream HPC Prep

- **Steps 1–5:** `Trimmomatic` → `Bowtie2` → `Picard` Dedup → `samtools` MAPQ ≥ 10 filter.
- **Steps 6–8:** Generates indexed BAMs (`06_index_bam/`), HOMER TagDirs & UCSC bedGraphs (`07_homer/`), and BigWigs (`08_bigwig/`).

Phase 3: Peak Calling & Merging

- **Step c:** Calls per-sample peaks via `MACS2` (from BAMs) and/or `HOMER` (from TagDirs).
- **Step d:** Filters reproducible peaks across replicates (`≥2`) and builds an inclusive union `<target>_master_peaks.bed` .

Phase 4: Downstream Discovery

- **Steps e–f:** HOMER gene/TSS annotation, normalized tag counts, and DESeq2 differential binding.
- **Steps g–i:** `deepTools` tornado heatmaps, HOMER/MEME motif discovery, and 5 automated QC plot suites.

2. Dropbox Inputs & Phase 3: Peak Calling & Consensus Merging

Resolving Staged Upstream Tracks → Per-Sample Peak Calling (Step c) → Replicate & Master Peak Merging (Step d)

Dropbox Input Subfolder (`${params.input_dir}`)	Config Parameter / Glob	File Contents	Downstream Steps Fed
`06_index_bam/`	`input_bam = ".../06_index_bam/*.bam"`	Deduplicated, MAPQ-filtered `.bam` + `.bam.bai`	**Step c1:** `MACS2_CALLPEAK`
`07_homer/tag_dir/`	`input_tagdirs = ".../07_homer/tag_dir/*"`	Sample HOMER Tag Directories (`<sample_id>/`)	**Steps c2, e, f:** HOMER FindPeaks, AnnotatePeaks, DiffPeaks
`08_bigwig/{deepTools,UCSC}/`	`bigwig_source = "deepTools"` (or `"UCSC"`)	RPKM (`deepTools`) or UCSC (`bedGraphToBigWig`) `.bw`	**Step g:** `DEEPTOOLS_METAPEAK` tornado heatmaps

Step c1: MACS2 Peak Calling (`MACS2_CALLPEAK`)

- **Toggle:** `run_macs2_callpeak = true`
- **Command:** `macs2 callpeak -t <id>.bam -f BAMPE -g ce -q 0.01`
- **Narrow vs. Broad:** Set `macs2_broad = true` for spreading histone domains (e.g., `H3K27me3`) or `false` for sharp marks (`H3K4me3` , TFs).
- **Published:** `09_peaks/macs2/<id>_peaks.narrowPeak` , `_peaks.xls` , `_summits.bed`

Step c2: HOMER Peak Calling (`HOMER_FINDPEAKS`)

- **Toggle:** `run_homer_findpeaks = true`
- **Command:** `findPeaks <tagdir> -style factor -o <id>_peaks.txt && pos2bed.pl`
- **Style Selection:** Supports `homer_peak_style = "factor"` or `"histone"` .
- **Published:** `09_peaks/homer/<id>_peaks.txt` & `<id>_peaks.bed`

Step d1: Replicate Peak Filtering (`BEDTOOLS_MERGE_REPLICATES`)

- **Toggle:** `run_merge_replicate_peaks = true` (uses `active_peak_caller = "macs2"` or `"homer"`)
- **Logic:** Groups sample peak BEDs by `condition` , merges overlapping coordinates via `bedtools sort | bedtools merge -c 4 -o count_distinct` , and retains intervals present in `≥ min_replicate_overlap` replicates (default `2`).
- **Published:** `10_merged_peaks/replicates/<condition>_replicated_peaks.bed`

Step d2: Master Peak Union per Target (`BEDTOOLS_MERGE_MASTER`)

- **Toggle:** `run_merge_master_peaks = true`
- **Logic:** Automatically excludes `IgG` controls, groups condition peaks by `target` (e.g., all `H3K4me3` conditions together, all `H3K27me3` conditions together), and merges into an inclusive master coordinate list.
- **Published:** `10_merged_peaks/master/<target>_master_peaks.bed`

3. Phase 4: Peak Quantification, Differential Binding & Tornado Plots

Measuring Signal Across Master Peaks (Step e), Statistical Contrasts (Step f), and Metapeak Visualization (Step g)

Step e: Peak-by-Peak Quantification & Gene Annotation (`HOMER_ANNOTATEPEAKS`)

- **Toggle & Input:** `run_homer_annotatepeaks = true` | Combines `<target>_master_peaks.bed` + all target Tag Directories (`07_homer/tag_dir/*`).
- **Command:** `annotatePeaks.pl <target>_master_peaks.bed ce11 -size given -cpu 4 -d <sample1_tagdir> <sample2_tagdir> ... > <target>_annotated_peaks.txt`
- **Key Outputs in `11_peak_quantification/<target>_annotated_peaks.txt` :** Nearest gene name & description, `Distance to TSS` , genomic feature class (`Promoter-TSS` , `Exon` , `Intron` , `Intergenic`), and depth-normalized tag counts (reads per 10M) for every sample across the exact width of each master peak (`-size given`).
- **Optional Width Normalization:** Run `bin/normalize_homer_peaks.py` to append length-normalized `*_per_kb` columns when comparing peaks of different widths.

Step f: Differential Peak Calling (`HOMER_GETDIFFERENTIALPEAKS`)

- **Toggle:** `run_homer_diffpeaks = true`
- **Configured Contrasts (`params.diff_contrasts`):** Pairs treatment vs. background conditions per target (e.g., `EV_INT_H3K4me3` vs. `N2_H3K4me3`).
- **Command:** `getDifferentialPeaksReplicates.pl -t <treat_tagdirs> -b <bg_tagdirs> -p <target>_master_peaks.bed -genome ce11 -all`
- **Statistical Output:** Runs DESeq2 across biological replicates and outputs fold change, raw *p*-value, and Benjamini–Hochberg adjusted **FDR** for all master peaks.
- **Published:** `12_differential_peaks/<contrast_id>_differential_peaks.txt`

Step g: Metapeak & Tornado Heatmaps (`DEEPTOOLS_METAPEAK`)

- **Toggle:** `run_deeptools_metapeak = true`
- **Signal Source:** Automatically selects `08_bigwig/deepTools/*.bw` (RPKM) or `08_bigwig/UCSC/*.bw` via `params.bigwig_source` .
- **Stage 1 (`computeMatrix reference-point`):** Aligns all target BigWig tracks on `<target>_master_peaks.bed` centers (`--referencePoint center -b 2000 -a 2000`).
- **Stage 2 (`plotHeatmap`):** Renders top summary profile curves and bottom per-peak sorted heatmaps (`--colorMap inferno`).
- **Published:** `13_metapeak_plots/<target>_metapeak_heatmap.pdf` & `_matrix.mat.gz`

4. Phase 4: Motif Discovery, Automated QC Plots & Pipeline Execution

Enriched Sequence Motifs (Step h), Publication Summary Figures (Step i), and One-Command Execution

Step h1: HOMER Motif Discovery (`HOMER_FINDMOTIFSGENOME`)

- **Toggle:** `run_homer_findmotifs = true`
- **Command:** `findMotifsGenome.pl <target>_master_peaks.bed ce11 <outdir> -size 100`
- **Outputs:** `14_motifs/homer/<target>_homer_motifs/` containing interactive web reports for **known TF motifs** (`knownResults.html`) and ***de novo* motifs** (`homerResults.html`).

Step h2: MEME Suite STREME Motifs (`MEME_STREME`)

- **Toggle:** `run_meme_streme = true`
- **Workflow:** Finds peak centers (`awk`) → expands ±50 bp windows (`bedtools slop -l 50 -r 49`) → extracts FASTA (`bed2fasta`) → runs `streme --nmotifs 20 --dna` .
- **Outputs:** `14_motifs/meme_streme/<target>_streme_motifs/streme.html`

Step i: Automated QC & Biological Summary Suite (`run_summary_plots = true` → `15_summary_plots/`)

- **`01_peak_counts/` (`PLOT_PEAK_COUNTS`):** Bar chart of per-sample MACS2/HOMER peaks, replicate-filtered peaks, and master union peak totals.
- **`02_genomic_features/` (`PLOT_ANNOTATED_SUMMARY`):** Stacked bar chart, donut chart, and TSS distance histogram (`Promoter-TSS` , `Exon` , `Intron` , `Intergenic`).
- **`03_correlation_and_pca/` (`PLOT_ANNOTATED_SUMMARY`):** Sample-to-sample Pearson correlation heatmap and PCA scatter plot on $\log_2$(Tag Count + 1).
- **`04_signal_distributions/` (`PLOT_ANNOTATED_SUMMARY`):** Violin/boxplots of raw tag counts vs. width-normalized density (tags/kb) and peak width histograms.
- **`05_differential_plots/` (`PLOT_DIFFERENTIAL_VOLCANO_MA`):** Volcano ($-\log_{10}$FDR vs. $\log_2$FC) and MA plots with automated top differential gene labels.

Quick Start: Running & Customizing from `Analysis_Run/`

```
# 1. Configure Analysis_Run/config/experiment.config (experiment = "cut_run103026A", bigwig_source = "deepTools")
# 2. Launch pipeline (automatically mounts Dropbox & resumes cached tasks):
cd Analysis_Run && ./run.sh
```